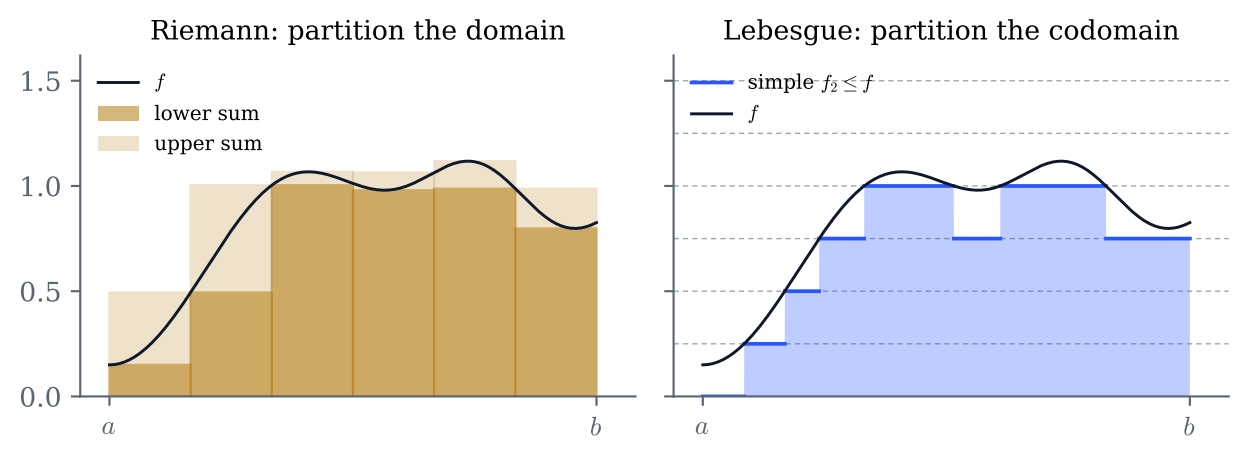

Riemann: partition the domain
f
lower sum
upper sum
1.5
1.0
0.5
0.0
a
b
Lebesgue: partition the codomain
simple $f_2 \le f$
f
a
b